

The Vector Is the Product: One-Pass Readouts of Frozen Language-Model Residuals Match the Autoregressive Loop

Abstract

On tasks whose answer is representable in a single forward pass, the autoregressive loop is not the only readout of a language model’s computation — and, we show, not a privileged one. We fit small supervised heads to the frozen final-token residual of six open-weight models (Qwen3-0.6B/4B/8B, Mistral-7B, Granite-3.1-8B, DeepSeek-V4-Flash) and run the readout head-to-head against a fairly-conditioned autoregressive loop — full context, balanced few-shot, scored by next-token log-probability over the answer set — matched on information access, scoring protocol, and supervision budget. On BoolQ (full validation), RuleTaker (n2k), and ARC-Challenge (full test), the one-pass readout **matches or beats** the fair loop at nearly every scale: it wins significantly where the model is weakest (RuleTaker +9.3pt at 0.6B, +13.3pt on Granite-8B; BoolQ +3.8pt at 0.6B against a 4×-budget loop), ties at mid scale, and loses decisively only in two regimes — small models on parametric knowledge (ARC) and a frontier MoE on serial deduction. The result survives label-shuffle nulls, random-projection selectivity, per-length deconfounds, and paired significance testing on identical rows. Sweeping the readout tap across depth, the verdict representation is fully assembled by mid-depth: the best mid-layer tap beats the final layer significantly at 9 of 18 task×model cells and never loses. The model, in short, knows more than it can say — on this task class, **the vector already has the answer**. We bound the claim explicitly: it is readout-vs-loop on a frozen backbone, not readout-vs-fine-tuned-specialist, and it does not exhibit a regime where a frozen loop is required.

1 Introduction

Autoregressive decoding is the universal serving interface for language models, but it is only one readout of the computation. Every generated token is produced by applying a fixed, next-token-trained unembedding to a residual vector; the answer a model *would* give is thus a particular linear read of a vector that exists, in full, after a single forward pass. This paper asks a sharp, falsifiable question about that vector: **on tasks whose answer can be represented in one forward pass, does a small readout head fit to the frozen residual recover the answer — and does the autoregressive loop**

add anything such a readout lacks?

The question is easy to answer sloppily. A probe that beats a zero-shot prompt has shown little: zero-shot understates the loop, and a token-probability decision rule is a known-weak baseline (Cho et al., 2025). A probe compared only to the model’s own self-report has shown something else entirely — that beliefs are present, not that a readout can *replace* the loop at producing the right answer. And any probe result without label-shuffle nulls, random-projection selectivity, and significance testing is one confound away from collapse. Our contribution is therefore as much a **discipline of controls** as a result: we run the comparison the probing literature has stopped short of — frozen open weights, a readout fit to gold labels, against a fairly-conditioned loop, matched on supervision budget, information, and scoring.

Concurrent work. The observation that activations carry content the response does not report is now independently published: Hazenoot et al. (2026) show a linear probe on frozen activations beats the same model’s own prompted yes/no answer on ESG concept measurement (11/12 comparisons), with the winning layer mid-network. Their loop baseline is zero-shot prompting — which we show understates the loop (§3, Ext 13) — and the comparison carries no paired significance or selectivity controls. We take the convergence as evidence the premise is ripe; the measurement we run here is the part that was missing.

Findings. On BoolQ (full validation, 3270 rows), RuleTaker (n2k pilot), and ARC-Challenge (full test), across six models from 0.6B to a frontier MoE:

1. **Parity or better almost everywhere.** The one-pass readout matches or beats the fair few-shot loop at nearly every scale — winning significantly exactly where the loop is weakest, and tying at mid scale (§3, §4).
2. **The loop wins in two bounded regimes only:** small models on parametric knowledge (ARC-Challenge at ≤ 7 B scale), and a frontier MoE on formal serial deduction (RuleTaker at DeepSeek-V4). Both gaps close or invert with scale and task type (§5).
3. **The answer is assembled by mid-depth.** Sweeping the readout tap across the residual stream, the best mid-layer tap beats the final layer significantly at 9/18 cells and never loses — the final layer is a *lower bound* on what the residual readout can do (§4.4).

4. **The margins can be large.** Where the loop’s verbal channel is weakest, the readout’s lead is largest: +13.3pt on RuleTaker with Granite-3.1-8B (readout 0.828 vs loop 0.695, McNemar $p = 1.3\text{e}-06$).

Why it matters. Later work on residual heads, latent taps, and efficient inference can treat “the answer is already in the vector” as established on a defined task class — not as a serving system we ship here, and not as a claim about all tasks. We state the boundaries explicitly (§5, §6): fine-tuned specialist encoders beat our readout; serial-depth tasks may require the loop; the frozen continuous-latent loop drifts. Within those bounds, the footnote is earned.

Scope. A companion position paper explores a serving architecture of kilobyte-scale readout heads (Paper 2). This paper contains no architecture claims and no production system.

2 Setup and Method

2.1 Two readouts of the same vector

One-pass readout. Let $h_T \in \mathbb{R}^d$ be the final-token residual of a frozen model after one forward pass over the task prompt (question + passage, truncated at `pad_max = 384`). We fit a head $f(h_T) \rightarrow \hat{y}$: either multinomial logistic regression or a one-hidden-layer MLP (width 128, 4-seed vote), both on standardized features. The head sees only task labels; the backbone is never updated.

Fair loop. The autoregressive baseline is *not* greedy decode — it is the strongest classifier the loop provides. The model receives the full context (balanced few-shot exemplars, `loop_pad_max 2048–8192`) and is scored by next-token log-probability over the closed answer set (Yes/No, or A–D). We report `loop.0` (zero-shot) and `loop.k` (k -shot); where noted we sweep k up to 64 with a $4 \times$ pad budget (Ext 13).

2.2 Matching requirements

A readout-vs-loop comparison is only meaningful if neither side is handicapped. We match on three axes: **information access** (both see the same question and passage; the loop additionally sees k exemplars — an advantage we grant it), **scoring protocol** (both produce a distribution over the same closed answer set; accuracy on identical rows), and **super-vision budget** (Ext 13 sweeps the loop’s exemplar budget to $4 \times$ the readout’s training budget at 0.6B scale).

2.3 Controls

Every headline number carries four controls:

- **Label-shuffle null** (`mlp.shuf1`): the head refit on permuted labels must sit at chance, ruling out leakage and spurious separability.

- **Random-projection selectivity** (`randproj`): a head on random Gaussian projections of the residual. If it retains accuracy, the signal is *diffuse* — no privileged subspace — which is what we find ($\text{perm} \approx \text{max} \gg \text{noise}$) at every scale.
- **Final-token vs full-context** (`ctx.mlp`): a mean-pooled full-context readout must underperform the final-token readout, confirming the verdict is *computed into* the final state rather than read off the passage surface.
- **Paired significance:** McNemar’s exact test and paired bootstrap on per-row predictions from identical val rows (`rowpreds_stats.py`), not pooled-accuracy eyeballing.

2.4 Models and artifacts

Six open-weight models spanning three families and a width ladder: Qwen3-0.6B (28×1024), Qwen3-4B (36×2560), Qwen3-8B (36×4096), Mistral-7B (32×4096), Granite-3.1-8B (40×4096), DeepSeek-V4-Flash (43×4096 , FP8 MoE). All runs are batch 8 on model-pinned GPUs (0.6B/4B H200, 8B/Mistral/Granite B200, DSV4 B300). Trained heads are persisted per layer as versioned artifacts (`heads/head_1{L}.npz`, $\sim 0.5\text{--}8.4$ MB); per-row predictions persist for paired tests. Harness: `bench.py`; run manifests embed the batch tag in cache keys after the Ext 15 cache-identity incident.

3 BoolQ: one-pass readout vs the fair loop

BoolQ is binary reading comprehension with the gold answer grounded in a passage — the cleanest case of a one-pass-representable task. We use the full validation split (9427 train / 3270 val).

3.1 Headline

Across all six models, the one-pass readout **wins at 0.6B, ties at 4B/8B and the two cross-family 8B models, and trails the $5 \times$ -context few-shot loop by $\sim 1\text{pt}$ only at the frontier MoE** (Table 1). At no scale does the fairly-conditioned loop *significantly* beat the readout on BoolQ.

3.2 Supervision asymmetry is ruled out (Ext 13)

The natural objection: the head sees thousands of labels, the loop sees k exemplars. We gave the loop its budget back — k swept to 64 balanced exemplars at $4 \times$ pad (8192) — at 0.6B/4B/8B. The loop’s k -curve **plateaus by $k=32$** (`loop.64` – `loop.32` = -0.0003 , $p = 1.00$ at 0.6B) and **never passes the readout where the readout was winning:** at 0.6B the

readout’s +3.8pt over the best loop stays significant ($p = 2.4\text{e-}05$); at 4B/8B all cells are exact ties. Supervision starvation is not the explanation (Fig. 6).

3.3 Three readings

1. **Parity or better at every scale; the edge is largest where the model is weakest.** The readout’s only significant win is at 0.6B — the loop’s verbal channel is the bottleneck there, and the readout channel saturates before it.
2. **The signal is distributed, not localized.** Random-projection heads retain accuracy ($\text{perm} \approx \text{max} \gg \text{noise}$) at every scale; no privileged subspace carries the verdict (Table 4).
3. **The verdict concentrates in the final token.** Mean-pooled full-context readouts underperform the final-token readout everywhere ($\text{ctx.mlp} \ll \text{last.mlp}$) — the answer is computed into the final state, not read off the passage.

3.4 FLOPs, precisely

Readout and *scoring* loop are both a single context forward pass — FLOP-equal up to context length. The readout’s structural lever is against a *decoding* loop — and here the non-self-termination is **measured**, not assumed: a greedy decode from the “Answer:” prompt never stops (25/25 hit the 300-token cap at 0.6B; 7/8 hit the 1000-token cap at DeepSeek-V4; `results/decode_baseline.json`). The parametric win is therefore real and unconditional: **a head instead of a decode loop — no new tokens, no KV-cache**. (We quantify the economics and their boundary against early-exit serving in Table 3; the decode row is measured.)

4 RuleTaker and ARC: serial depth and parametric knowledge

BoolQ is passage-grounded. We test the two ways a task can be harder for a one-pass readout: **serial deduction depth** (RuleTaker) and **parametric knowledge with no passage** (ARC-Challenge).

4.1 RuleTaker n2k: matched loop rows, per-depth strata

RuleTaker is closed-world rule reasoning with labeled deduction depth (0/1/2/3/5 + NatLang). Protocol: 2000 train / 1000 val, loop on 400 matched rows, $k \in \{0, 8\}$, batch 8. On Qwen, the matched one-pass MLP **beats the fair 8-shot loop at every scale** (+2 to +9pt; significant at 0.6B, $p = 3.7\text{e-}03$). The cross-family models sharpen the picture: Mistral-7B +8.5pt ($p = 0.017$), **Granite-3.1-8B** +13.3pt

($p = 1.3\text{e-}06$) — **the largest readout margin in the campaign**. The one clean loop win is DeepSeek-V4 (+7.4pt, $p = 1\text{e-}03$), driven by the shallow-depth strata.

Per-depth (same rows): no sharp “only the loop works past depth D ” cliff. Accuracy degrades gradually with depth; $d=5$ remains well above chance. The DSV4 loop win concentrates at shallow depths ($d0/d1$, +10–15pt); at 0.6B the loop *collapses* with depth ($d5$ loop.8 0.392 $\ll$ readout 0.544). Thin bins ($n = 21\text{--}51$) are reported with n and not over-read (Fig. 5).

4.2 ARC-Challenge: the loop’s best case

ARC-Challenge is parametric science knowledge with no passage — the answer must come from the weights, not the context. Here the loop wins at small scale: **−10.6pt at 0.6B, −4.6pt at 4B** (both $p < 1\text{e-}05$), narrowing to −0.5pt at 8B and −0.7pt at DSV4 (both ns). The cross-family models replicate the pattern: Mistral-7B −5.1pt ($p = 9.3\text{e-}05$), Granite-8B −8.2pt ($p = 5.8\text{e-}10$). The gap **closes monotonically with scale** — few-shot exemplars teach a task format the residual head must otherwise learn from labels, and the advantage shrinks as the parametric answer saturates the residual.

4.3 The task law

Across 18 task×model cells, a single law organizes everything (Fig. 1): **the loop wins in exactly two regimes — small models on parametric knowledge (ARC), and a frontier MoE on formal serial deduction (RuleTaker at DSV4). Everywhere else the one-pass readout wins or ties**. The ARC crossover tracks *capability*, not family (Granite/Mistral sit where Qwen did at equal accuracy); the RuleTaker crossover tracks *serial structure at frontier scale*.

4.4 Readout placement: the final layer is a lower bound

Every headline number above taps the final residual layer. We swept the tap across depth (10 layers, paired McNemar vs the final-layer readout, trained heads persisted at every tap). The curve is the same everywhere: monotone rise, mid-depth plateau, slight terminal decline (Table 2, Fig. 2).

- **The final layer never wins.** Mid-depth taps beat the final layer significantly at **9/18 cells and never lose significantly** — the headline numbers were a lower bound on the readout, not its ceiling.
- **The optimum is task-stable, not scale-driven.** BoolQ/RuleTaker peak at 50–86% depth at every scale; RuleTaker’s optimum sits at $\sim 2/3$ depth from 0.6B through DSV4. ARC drifts to the final layer at $\geq 8B$ — once the parametric answer saturates the residual, late-layer specialization stops costing anything.

- **Placement converts parity into wins.** Where the loop catches the final-layer readout (BoolQ 8B: L35 ties loop), the best mid-layer tap re-establishes the lead (L30 +1.1pt, $p = 0.016$; on Granite, L20 +1.8pt, $p = 1e-03$).
- **Wide-shallow plateaus earlier.** Mistral (32×4096) onsets at 35–58% depth vs Qwen3-8B’s uniform 69% at the same width — at fixed width, the shallower stack assembles the verdict sooner. Granite (40×4096 , deep) sits between the two.

Causal hedge. The monotone-rise $\rightarrow$ plateau $\rightarrow$ terminal-decline shape is *consistent with* late-layer specialization toward next-token surface form, but we do not demonstrate it: we never ablated deep layers against generation, and never probed next-token structure per layer. The direct test (a per-layer next-token probe) is future work; “the verdict readout needs less than full depth” and “the final third is load-bearing for decoding” are different claims, and only the first is on the table here.

5 The boundary: where the loop could win, and why it doesn’t cleanly here

A single forward pass is a bounded-depth computation (TC^0 in the circuit-complexity framing); the loop’s advantage should appear exactly where serial depth is irreducible. We hunted that regime and found it only weakly exhibitable on frozen base weights:

- **RuleTaker per-depth** shows a gentle slope, not a regime switch — the loop edges the readout by a few points at $d=5$ (noisy thin bins), and the one decisive loop win (DSV4) concentrates at *shallow* depths, where the frontier model’s in-context deduction is strong.
- **ARC** shows the loop’s real advantage is *parametric retrieval under a learned task format*, not serial compute — and it closes with scale.
- **Synthetic mechanism** (multi-hop transitivity with shuffled facts; Fig. 7): the relation is linearly decodable from the frozen residual (0.93–0.97) where the zero-shot decoder sits at chance; a 2-demo budget surfaces the loop’s inference but never exceeds the one-pass MLP. The measured limit is **order-locked binding**, not depth.

The remaining “loop wins” cell — a TC^0 -hard task on which a *frozen* loop decisively beats one-pass — is largely unoccupied in our sweep; the counter-position (COCONUT, looped-latent) buys serial depth by *training*, which is out of scope here.

6 Discussion

The thesis, in its conditional form. On closed-form, one-pass-representable tasks, a small head on the frozen residual matches a fair autoregressive classifier — so the answer is already in the vector. This is a measurement, not a metaphysics: we do not claim the loop never computes, and we identify the two regimes where it does (§5).

The model knows more than it can say. The readout’s largest margins are exactly where the loop is weakest (Granite RuleTaker +13.3pt; 0.6B everywhere) — the readout channel saturates before the verbal channel. This is the internal-knowledge/external-behavior disconnect Orgad et al. (2025) document from the interpretability side; we show it has *operational* content: the readout is a drop-in replacement for the classifier the loop provides.

Separability of readout from generative interface. The unembedding is one readout of the residual — a fixed, next-token-trained one. That a supervised head matches it is the default expectation once stated that way; the empirical content is *where* it fails (ARC at scale, DSV4 deduction) and that the failure modes are orderly.

Specialist frontier (acknowledged). Fine-tuned encoder classifiers (RoBERTa-large ~ 0.87 , DeBERTa-v3 ~ 0.88 , T5-11B 0.91 on BoolQ) **beat** our 8B one-pass readout (0.879). Our claim is readout-vs-loop on a frozen backbone — the comparison a serving stack faces when the weights are already loaded — not readout-vs-fine-tuned-specialist. Where a specialist can be trained and shipped, it wins on raw accuracy; the readout’s economics are marginal-cost (the forward pass already happened).

Methodological corrections as worked example. This campaign corrected itself twice in ways worth recording: a loop-score cache keyed without batch identity silently inflated historical loop numbers (Ext 15 — found by signature: pre-fix runs agreed with each other, post-fix runs agreed with each other, readouts never affected); and supervision asymmetry, which we closed by giving the loop a $4 \times$ exemplar budget (Ext 13). Both corrections *sharpened* the headline.

What we are not claiming. A frozen continuous latent loop works (it drifts — Exp 7); the loop never computes; we beat fine-tuned specialists; we shipped residual-first inference. Paper 2 explores the serving architecture as a proposal.

7 Limitations

- **RuleTaker is an n2k pilot** (2000/1000, natural depth mix, $d3$ fat), not the full test; loop rows are 400.
- **Paired tests cover all 18 placement cells and the BoolQ budget arm**, but the per-depth loop strata are thin ($n = 21-51$).
- **Supervision asymmetry is bounded, not eliminated:** the head still sees more labels than the loop’s exemplars

at scale parity (Ext 13 closes it at 0.6B; Arm C — a fine-tuned decoder — is future work).

- **Reproducibility:** per-layer heads are persisted for all sweep runs; the two oldest canonical headline runs (Ext 8/9) predate head persistence. The FLOP-fraction figure is order-of-magnitude (active-param counts for the MoE are unmeasured — Table 3).
- **Scope:** in-knowledge, closed-form tasks only. No claim transfers to open-ended generation.

generating tokens — and the autoregressive loop adds nothing such a readout lacks. The claim is conditional, bounded, and control-hardened: it holds across three tasks, six models, and three families; it fails in two orderly regimes; and its margins are largest where the loop is weakest. **The vector already has the answer.** The construction of serving machinery on that substrate is companion work; here, we measured the substrate.

8 Related Work

Accessing what a frozen model knows is established from three directions, none of which runs our comparison. *Behavioral* work (Turpin et al., 2023) shows CoT rationales can be unfaithful — text-only access, no representation. *Latent-belief* work reads the representation directly: CCS (Burns et al., 2022), pre-CoT belief probes (Cox et al., 2026), belief-formation timing (Boppana et al., 2026), and the correctness/self-knowledge cluster (Kadavath et al., 2022; Orgad et al., 2025; “No Answer Needed”, 2026) — all predict a property of the model’s own output, none against gold labels, none against the fair loop. *Decoding-suboptimality* work (Cho et al., 2025; Buckmann & Hill, 2024; and closest, Hazenoot et al., 2026) shows hidden-state readouts beat the token-probability decision rule or zero-shot prompting — baselines their own analyses show are weak. *Probe-for-verification* work (ReProbe, Ni et al., 2026) uses internal states to improve generation, not replace it.

The empty cell — frozen open weights, a readout fit to gold labels, run head-to-head against a fairly-conditioned loop, matched on supervision budget, information, and scoring, under format/shuffle/randproj controls — is the one this paper occupies. Turpin saw behavior, Cox saw belief, Orgad read its correctness, Cho beat a baseline they showed was weak, Hazenoot beat zero-shot prompting, ReProbe verified steps to generate better; we compare the readout and the loop at producing the right answer, because the weights are open and frozen.

Delimiting counter-position. COCONUT (Hao et al., 2024) and looped-latent transformers (Saunshi et al., 2025) *train* a continuous latent loop to buy serial depth — they set the boundary we test without training. Early-exit serving (EE-LLM, CALM, LayerSkip, layer-removal) operationalizes placement redundancy into a serving optimization; it shares our premise and is prior art for any “skip the tail” claim, which we do not make.

9 Conclusion

On closed-form, one-pass-representable tasks, a small readout head fit to the frozen residual recovers the answer without

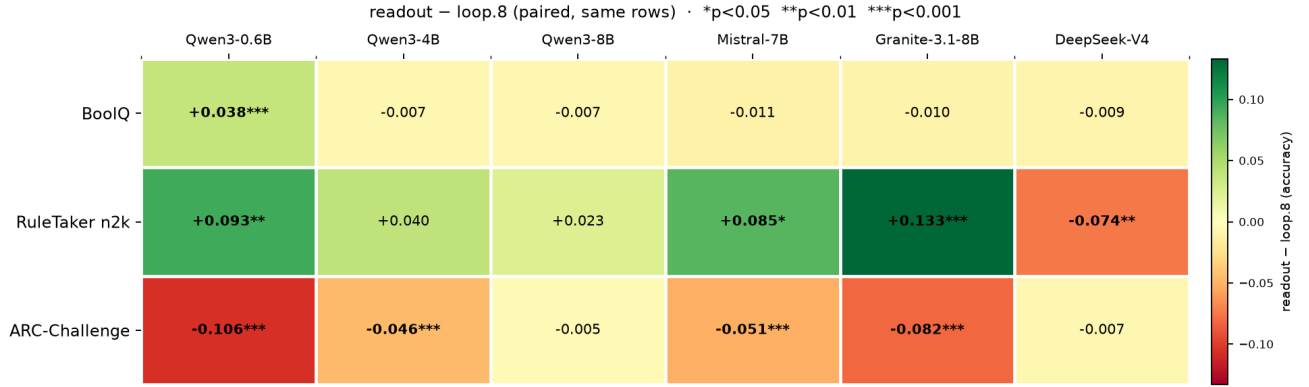


Figure 1: The task law. readout — loop. k (paired, same rows) per task×model cell, with McNemar significance. Green everywhere except two red corners — small models on parametric ARC, and the frontier MoE on serial RuleTaker.

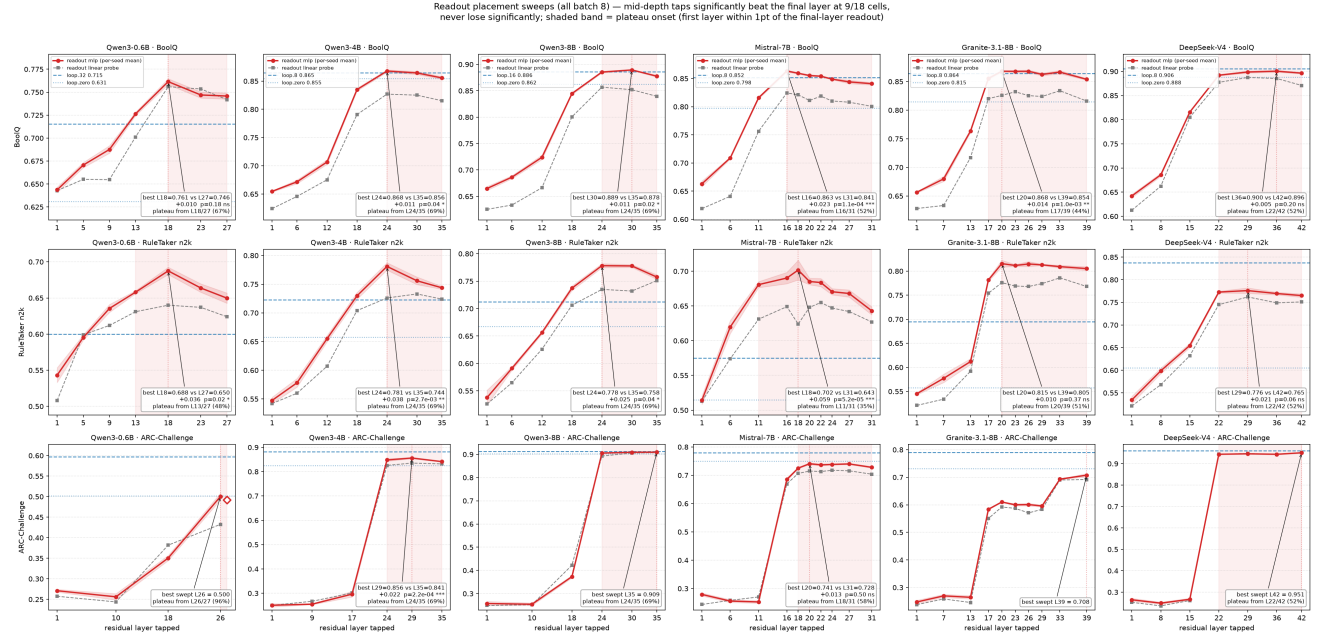


Figure 2: Readout probe-placement sweeps (18 cells). One-pass readout accuracy vs tapped residual layer; loop arms as horizontal references; paired McNemar annotation per panel. Mid-depth taps win significantly at 9/18 cells and never lose.

Table 1: One-pass residual readout vs. fairly-conditioned autoregressive loop (accuracy). Readout = small MLP on the frozen final residual (4-seed vote); loop = next-token log-prob over the closed answer set, full context, balanced few-shot. Δ = readout − loop.k, paired McNemar on identical rows ($*p < 0.05$, $**p < 0.01$, $***p < 0.001$). All runs batch 8; Qwen3 0.6B/4B on H200, Qwen3-8B/Mistral-7B/Granite-3.1-8B on B200, DeepSeek-V4-Flash on B300. ^a BoolQ loops: Qwen at pad 8192 with best $k \in \{0, \dots, 64\}$, Mistral/Granite/DSV4 at pad 2048, $k=8$. ^b RuleTaker n2k: readout scored on the loop’s exact 400 rows. ^c ARC-Challenge: full test (1165 rows), $k=8$, pad 2048.

Task	Model	Readout	loop.0	loop.k	Δ	p (McNemar)
BoolQ ^a	Qwen3-0.6B	0.753	0.631	0.715	+0.038***	2.4e−05
	Qwen3-4B	0.862	0.854	0.869	−0.007	0.22
	Qwen3-8B	0.879	0.862	0.886	−0.007	0.24
	Mistral-7B	0.841	0.798	0.852	−0.011	0.21
	Granite-3.1-8B	0.854	0.815	0.864	−0.010	0.64
	DeepSeek-V4-Flash	0.896	0.888	0.906	−0.009	0.39
RuleTaker n2k ^b	Qwen3-0.6B	0.638	0.600	0.545	+0.093**	3.7e−03
	Qwen3-4B	0.738	0.653	0.698	+0.040	0.20
	Qwen3-8B	0.753	0.675	0.730	+0.023	0.45
	Mistral-7B	0.660	0.515	0.575	+0.085*	0.017
	Granite-3.1-8B	0.828	0.558	0.695	+0.133***	1.3e−06
	DeepSeek-V4-Flash	0.763	0.605	0.838	−0.074**	1e−03
ARC-Challenge ^c	Qwen3-0.6B	0.492	0.502	0.597	−0.106***	3.2e−11
	Qwen3-4B	0.837	0.826	0.882	−0.046***	2.9e−06
	Qwen3-8B	0.908	0.902	0.913	−0.005	0.57
	Mistral-7B	0.728	0.750	0.779	−0.051***	9.3e−05
	Granite-3.1-8B	0.708	0.732	0.790	−0.082***	5.8e−10
	DeepSeek-V4-Flash	0.951	0.961	0.959	−0.007	0.23

Table 2: Readout probe-placement sweeps (18 cells, all batch 8): best mid-depth tap vs the final-layer readout, paired McNemar on identical rows. *Layers* = stack depth (28/36/36/32/40/43); taps are residual-layer indices. *Plateau onset* = first swept layer within 1 pt of the final-layer readout — the verdict representation is fully assembled from there on; at DeepSeek-V4 this happens by L22/42 (52% depth) on *all three tasks*, with ARC showing the sharpest snap-in (near-chance at L15 → 0.944 at L22). Mid-depth taps win significantly at 9/18 cells and never lose significantly.

Task	Model	Layers	Best tap	Depth	Plateau onset	mlp (best)	mlp (final)	Δ	p
BoolQ	Qwen3-0.6B	28	L18	64%	L18 (67%)	0.761	0.746	+0.010	0.18
	Qwen3-4B	36	L24	67%	L24 (69%)	0.868	0.856	+0.011*	0.038
	Qwen3-8B	36	L30	83%	L24 (69%)	0.889	0.878	+0.011*	0.016
	Mistral-7B	32	L16	50%	L16 (52%)	0.863	0.841	+0.022***	1.1e−04
	Granite-3.1-8B	40	L20	51%	L17 (44%)	0.868	0.854	+0.018**	1.0e−03
	DeepSeek-V4-Flash	43	L36	86%	L22 (52%)	0.900	0.896	+0.005	0.20
RuleTaker n2k	Qwen3-0.6B	28	L18	64%	L13 (48%)	0.688	0.650	+0.036*	0.020
	Qwen3-4B	36	L24	67%	L24 (69%)	0.781	0.744	+0.038**	2.8e−03
	Qwen3-8B	36	L24	67%	L24 (69%)	0.778	0.758	+0.025*	0.040
	Mistral-7B	32	L18	56%	L11 (35%)	0.702	0.643	+0.059***	5.2e−05
	Granite-3.1-8B	40	L20	51%	L20 (51%)	0.816	0.805	+0.011	0.37
	DeepSeek-V4-Flash	43	L29	67%	L22 (52%)	0.776	0.765	+0.021	0.057
ARC-Challenge	Qwen3-0.6B	28	L26	93%	L26 (96%)	0.500	0.492	+0.013	0.082
	Qwen3-4B	36	L29	81%	L24 (69%)	0.856	0.841	+0.022***	2.2e−04
	Qwen3-8B	36	L35	97%	L24 (69%)	0.909	0.909	+0.001	1.00
	Mistral-7B	32	L20	63%	L18 (58%)	0.741	0.728	+0.013	0.50
	Granite-3.1-8B	40	L39	100%	L39 (100%)	0.708	0.708	0.000	—
	DeepSeek-V4-Flash	43	L42	98%	L22 (52%)	0.951	0.951	−0.005	0.29

Table 3: What each access to the frozen residual costs (economics orientation, not a benchmark). Status stamps every row: “measured” = our run protocol; “projected” = forward arithmetic we did not run; “off-axis” = a real competing claim on a different cost axis (memory bandwidth), not operator FLOPs. Readout & scoring-loop are the same single pass at equal context — the readout’s structural gap is **no generation**, not layer-skipping. Early-exit serving shares our placement premise and is prior art for any “skip the tail” claim.

Access	Cost shape	Generates?	Status
readout (final tap)	1 fwd , short ctx	no	measured
scoring loop “.8”	1 fwd, long ctx	no	measured
early-exit decode	1 fwd, tail truncated	yes	projected / prior art
decode / output	K fwds (per token)	yes	measured (no self-stop)
Verity / bandwidth	1 fwd, memory-bound	yes	off-axis

Note. The decode row is measured: greedy decode from the Answer: prompt never self-terminates (25/25 hit the 300-token cap at 0.6B; 7/8 hit the 1000-token cap at DeepSeek-V4, `results/decode_baseline.json`).

Table 4: Controls, all 18 cells. Every headline number clears all four controls. *shuff*: head refit on permuted labels sits at chance. *randproj*: a head on random Gaussian projections retains accuracy with $\text{perm} \approx \max \gg \text{noise}$ — the signal is *diffuse*. *ctx*: a mean-pooled full-context readout underperforms the final-token readout (last.mlp) — the verdict is computed into the final state, not read off the passage. Chance accuracy is ≈ 0.62 (BoolQ), 0.50 (RuleTaker), 0.27 (ARC).

Task	Model	shuff (null)	randproj max	randproj perm	randproj noise	ctx.mlp	last.mlp
BoolQ	Qwen3-0.6B	0.533	0.734	0.725	0.601	0.667	0.746
	Qwen3-4B	0.542	0.861	0.859	0.604	0.717	0.856
	Qwen3-8B	0.528	0.874	0.874	0.607	0.739	0.878
	Mistral-7B	0.537	0.810	0.803	0.607	0.685	0.841
	Granite-3.1-8B	0.532	0.848	0.849	0.607	0.679	0.854
	DeepSeek-V4	0.522	0.889	0.891	0.602	0.678	0.896
RuleTaker	Qwen3-0.6B	0.511	0.635	0.635	0.489	0.517	0.650
	Qwen3-4B	0.474	0.718	0.726	0.505	0.586	0.744
	Qwen3-8B	0.488	0.736	0.745	0.496	0.575	0.758
	Mistral-7B	0.495	0.645	0.634	0.496	0.503	0.643
	Granite-3.1-8B	0.491	0.756	0.757	0.496	0.545	0.805
	DeepSeek-V4	0.500	0.726	0.716	0.506	0.518	0.765
ARC	Qwen3-0.6B	0.278	0.470	0.470	0.254	0.278	0.492
	Qwen3-4B	0.269	0.802	0.803	0.255	0.349	0.841
	Qwen3-8B	0.287	0.869	0.867	0.251	0.399	0.909
	Mistral-7B	0.259	0.679	0.678	0.251	0.277	0.728
	Granite-3.1-8B	0.284	0.598	0.588	0.251	0.279	0.708
	DeepSeek-V4	0.269	0.942	0.940	0.236	0.439	0.951

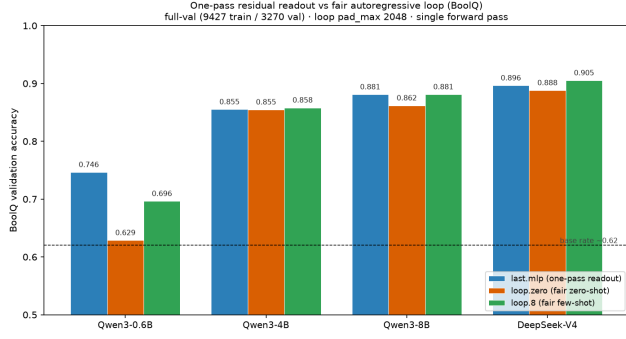


Figure 3: BoolQ: readout vs loop across four scales (full val).

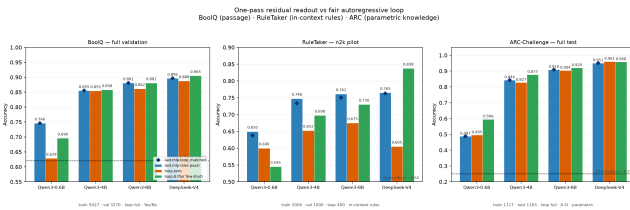


Figure 4: Three-task head-to-head (BoolQ · RuleTaker · ARC) across models.

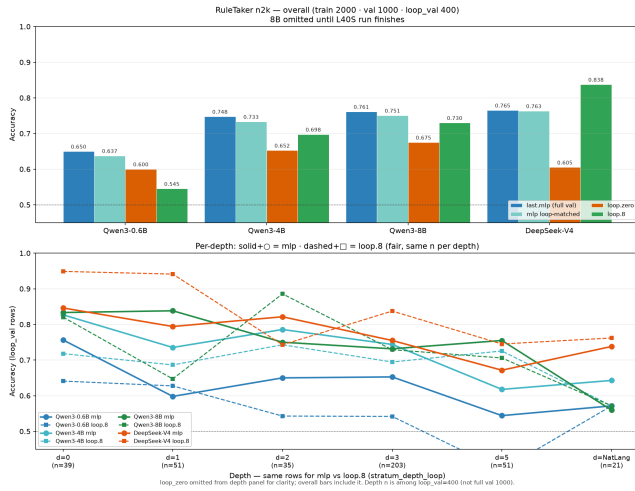


Figure 5: RuleTaker overall + per-depth strata, readout vs loop.

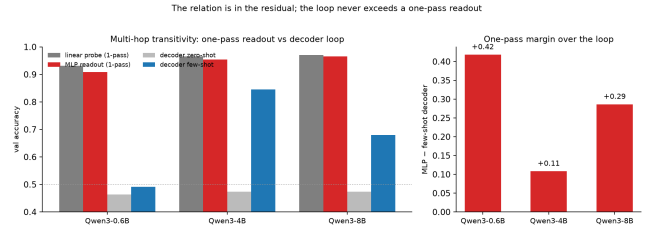


Figure 7: Mechanism (synthetic). Multi-hop transitivity: one-pass readout vs decoder loop across Qwen3 scale. The loop never exceeds a one-pass readout.

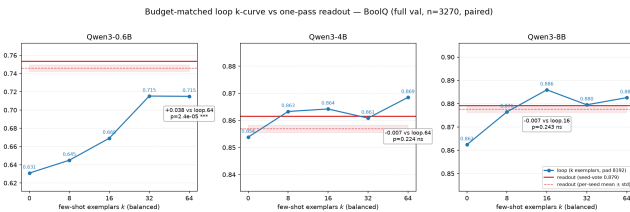


Figure 6: Supervision budget (Ext 13). The loop's k -curve plateaus by $k=32$ and never passes the readout.